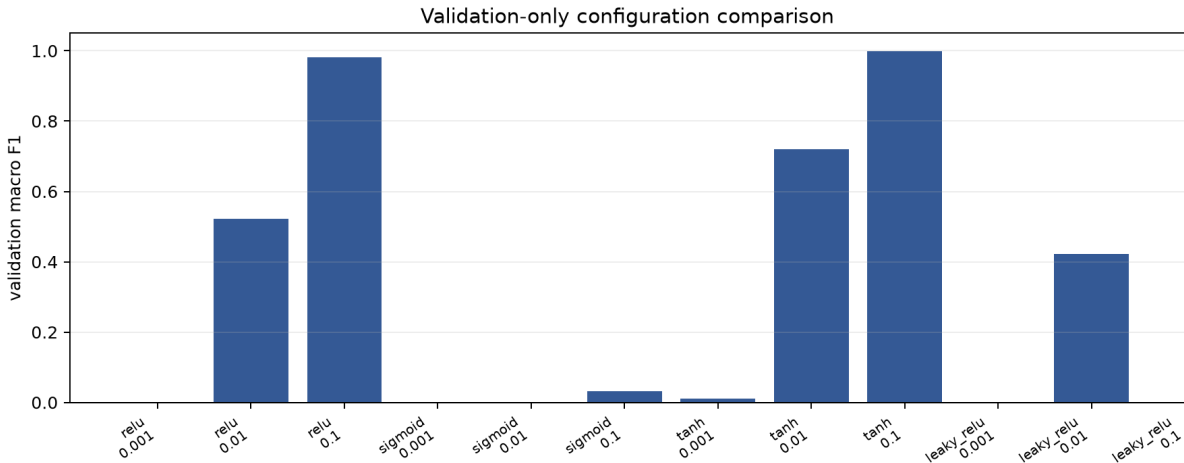


Fruits-360 CNN Experiment Report

Expanded 131-class activation and learning-rate study | Evidence-derived report

Executive summary

Twelve controlled CNN runs compared four hidden activations and three SGD learning rates. The validation-selected configuration was tanh at learning rate 0.1 (validation macro F1; accuracy tie-break; saved train-only checkpoint, no refit). Held-out test accuracy was 93.23%; macro F1 was 92.74%. Results are bounded by the recorded scope.



Provenance and scope

Dataset: Fruits-360 2020.05.18.0 / 131-class snapshot. Upstream commit: ffd2d14eada57a0c5537700190b309cfea2e120. Source: <https://github.com/Horea94/Fruit-Images-Dataset>. Snapshot counts: 67,692 Training and 22,688 Test images across 131 classes. Sampled counts, rather than upstream totals, govern this report.

Split	Images	Role
Training	54163	parameter updates
Validation	13526	selection
Held-out test	22688	final evaluation

Sampling: All single-fruit Training and Test JPGs; exact duplicates excluded. Seed 20260919; 8 epochs; batch size 128; input 32; optimizer SGD, momentum=0.

Methodology and formulas

The split, model architecture, initialization controls, batch size and epoch budget are held constant. Activation and learning rate are the only comparisons. Validation macro F1 selects a train-only checkpoint, with accuracy as tie-breaker. That checkpoint alone receives the recorded held-out test evaluation; no refit occurs. ReLU outputs $\max(0, x)$; Sigmoid maps to $(0, 1)$; Tanh maps to $(-1, 1)$; Leaky ReLU uses a negative slope of 0.1. Hidden activations change while the softmax output remains fixed.

Stage	Operation
Input	RGB image resized to 32
Architecture	Conv2D(8,3,same), activation, MaxPool2D, Conv2D(16,3,same), activation, MaxPool2D, Flatten, Dense(64), activation, Dense(131,softmax)
Classifier	Softmax over 131 classes
Loss / optimizer	Sparse categorical cross-entropy / SGD

Softmax divides $\exp(\text{class score})$ by the sum of exponentiated scores. Cross-entropy is the mean negative log probability of the true class; SGD: $\text{weights_new} = \text{weights_old} - \text{learning_rate} * \text{gradient}$. Precision = $\text{TP} / (\text{TP} + \text{FP})$; recall = $\text{TP} / (\text{TP} + \text{FN})$; F1 = $2\text{PR} / (\text{P} + \text{R})$. Macro values average classes equally.

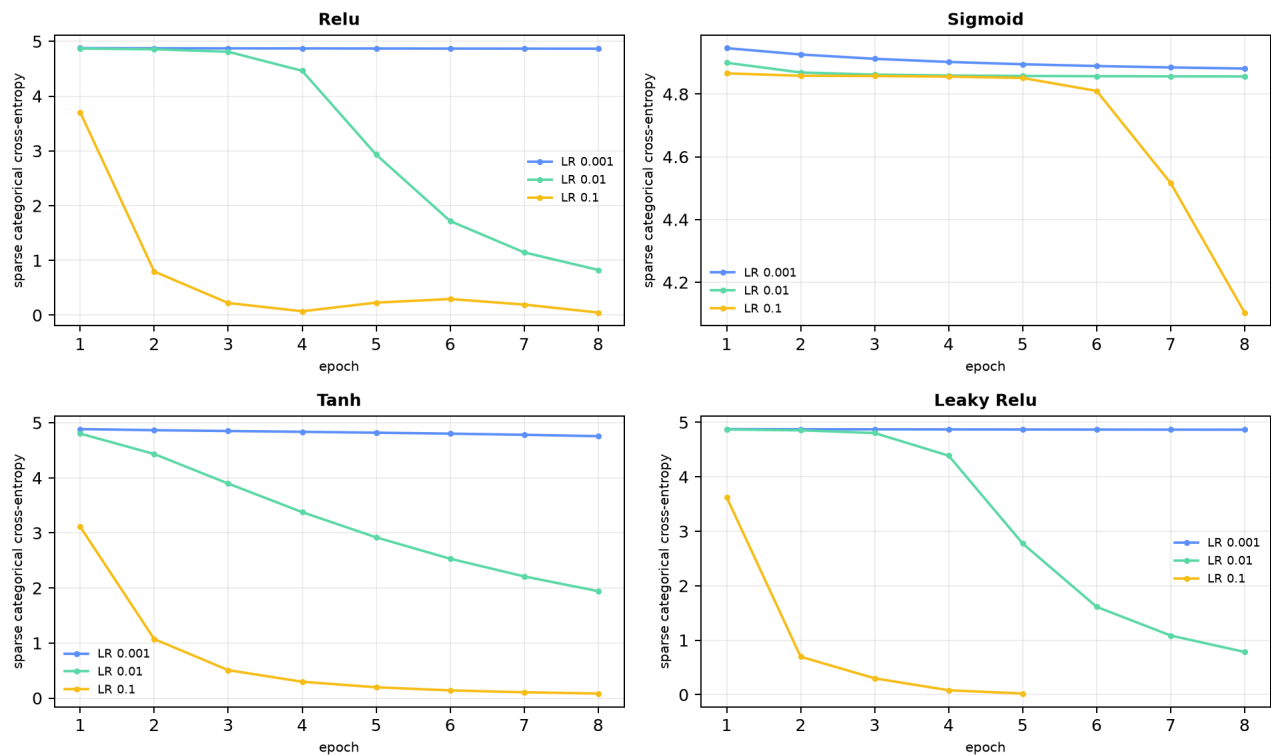
All 12 validation comparisons

Activation	LR	Acc	Macro P	Macro R	Macro F1	sec
relu	0.001	0.0109	0.0001	0.0076	0.0002	103.2
relu	0.01	0.5302	0.6740	0.5164	0.5214	107.6
relu	0.1	0.9847	0.9844	0.9828	0.9810	97.2
sigmoid	0.001	0.0109	0.0001	0.0076	0.0002	104.3
sigmoid	0.01	0.0146	0.0001	0.0076	0.0002	103.8
sigmoid	0.1	0.1008	0.0328	0.0790	0.0320	102.5
tanh	0.001	0.0470	0.0084	0.0337	0.0113	102.6
tanh	0.01	0.7536	0.7777	0.7369	0.7193	102.3
tanh	0.1	0.9979	0.9982	0.9978	0.9979	101.7
leaky_relu	0.001	0.0113	0.0002	0.0081	0.0003	103.4
leaky_relu	0.01	0.4483	0.6265	0.4448	0.4227	125.9
leaky_relu	0.1	N/A (diverged)	N/A (diverged)	N/A (diverged)	N/A (diverged)	108.9

Every figure in this table is validation-only. Leaky ReLU at 0.1 diverged from epoch 6: non-finite loss makes its final metrics unavailable, not zero. It is excluded from selection.

Training loss — all activation curves

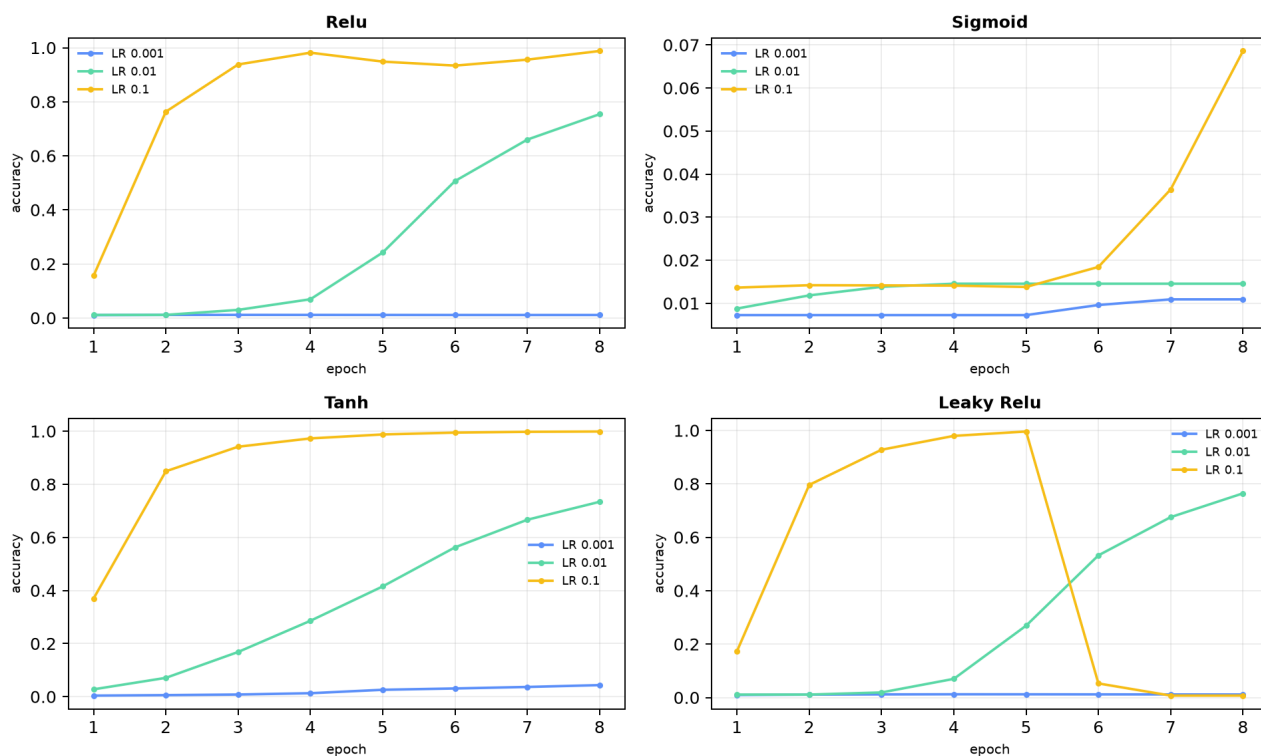
Training loss curves — all twelve measured runs



Four panels show all learning rates. Missing Leaky ReLU 0.1 loss points from epoch 6 indicate numerical divergence, not missing measurements. Accuracy from invalid outputs is not meaningful.

Training accuracy — all activation curves

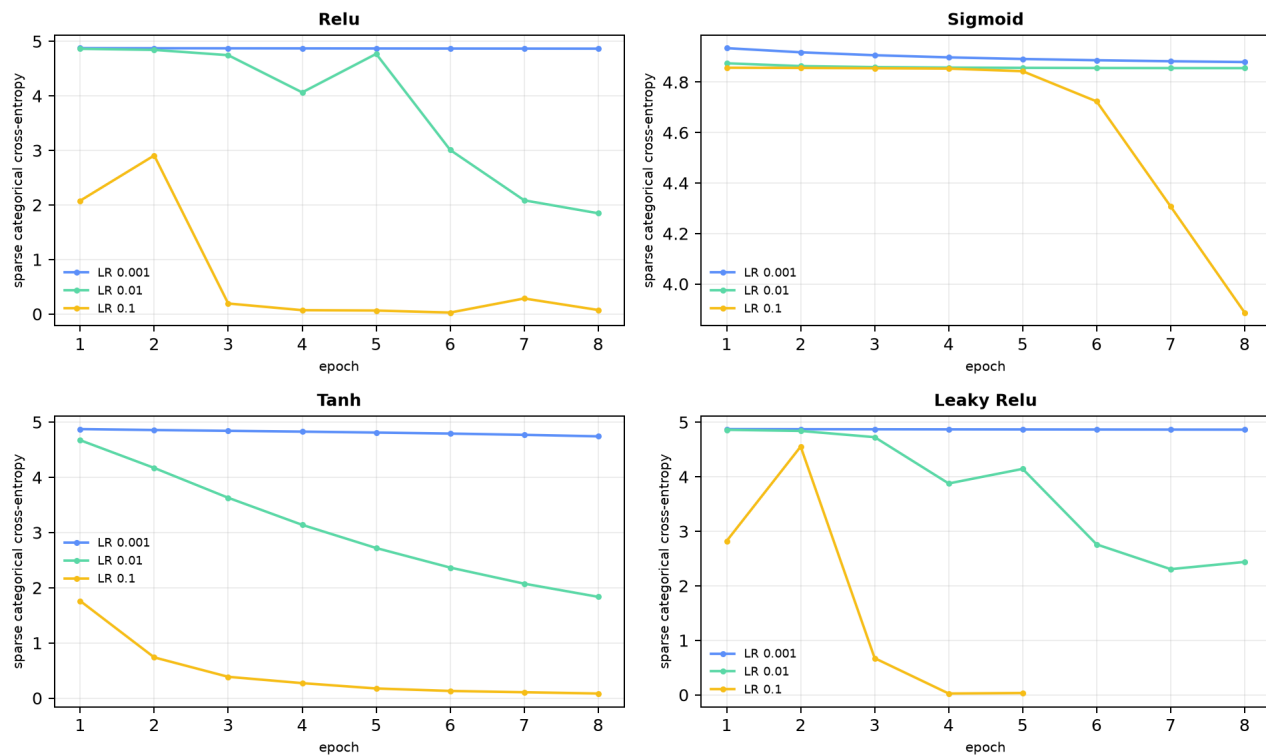
Training accuracy curves — all twelve measured runs



Four panels show all learning rates. Missing Leaky ReLU 0.1 loss points from epoch 6 indicate numerical divergence, not missing measurements. Accuracy from invalid outputs is not meaningful.

Validation loss — all activation curves

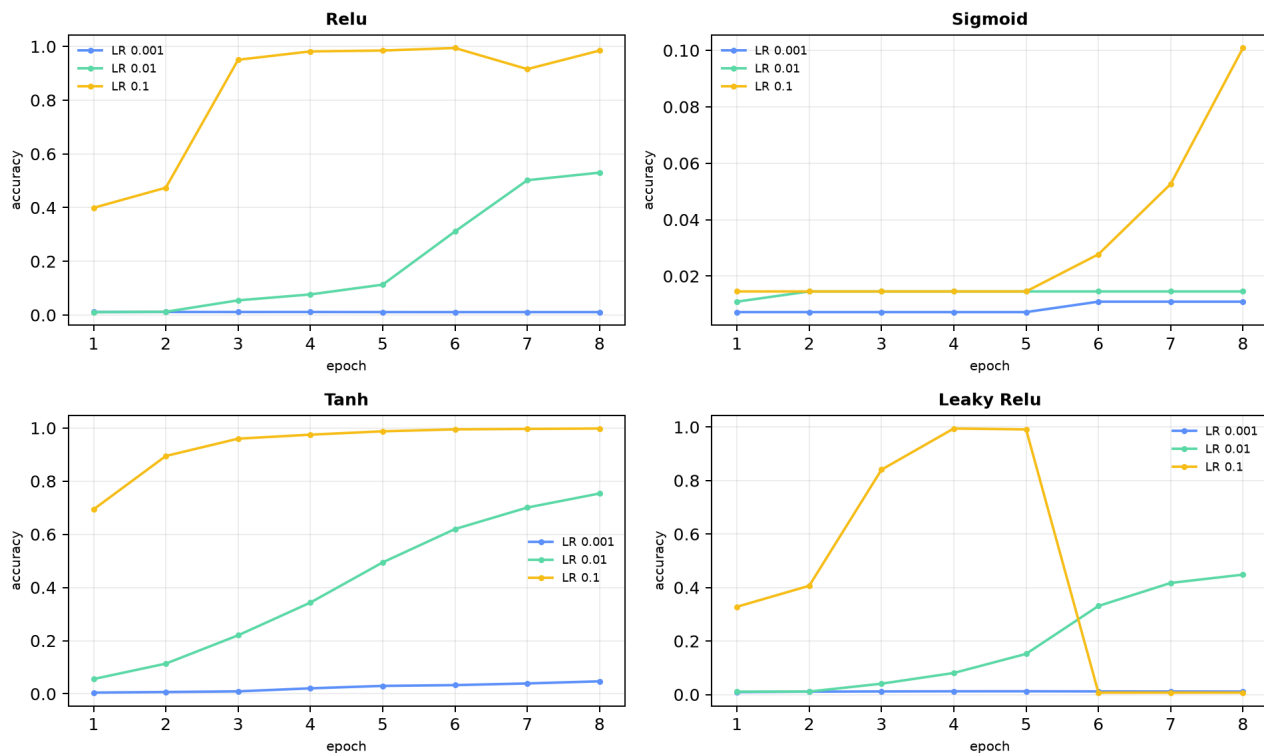
Validation loss curves — all twelve measured runs



Four panels show all learning rates. Missing Leaky ReLU 0.1 loss points from epoch 6 indicate numerical divergence, not missing measurements. Accuracy from invalid outputs is not meaningful.

Validation accuracy — all activation curves

Validation accuracy curves — all twelve measured runs



Four panels show all learning rates. Missing Leaky ReLU 0.1 loss points from epoch 6 indicate numerical divergence, not missing measurements. Accuracy from invalid outputs is not meaningful.

Held-out test evaluation

Metric	Measured value
accuracy	0.9323
precision macro	0.9406
recall macro	0.9281
f1 macro	0.9274

Summary of classes with the lowest diagonal counts. Detailed per-class measurements are exported separately as `per_class_results.csv`.

Class	Correct	Errors	Most common other prediction
Corn	54	96	Physalis with Husk
Beetroot	63	87	Pear Stone
Pear Kaiser	75	27	Plum 3
Mangostan	79	23	Chestnut
Potato Sweet	84	66	Potato Red
Ginger Root	91	8	Pear Stone
Eggplant	95	61	Pepper Green
Apple Braeburn	97	67	Nectarine
Pear Monster	103	63	Physalis with Husk
Tomato not Ripened	103	55	Pepper Green
Pomegranate	104	60	Strawberry Wedge
Apple Crimson Snow	109	39	Apple Pink Lady
Cucumber Ripe	113	17	Apple Red Delicious
Peach	113	51	Grape Pink
Potato Red Washed	114	37	Onion Red
Pear	115	49	Apple Golden 3
Nectarine	115	49	Apple Braeburn
Pepino	117	49	Pear Williams

Limitations

The claim is limited to the saved sampling, preprocessing, fixed seed, short training budget and source distribution. This evidence does not establish external-camera, out-of-distribution, repeated-seed, calibrated or deployment performance.

Discussion and conclusion

Interpretation: Tanh at 0.1 reached validation macro F1 0.9979 and held-out test macro F1 0.9274. The validation-to-test gap shows that very high validation performance is not a guarantee of equivalent unseen-test accuracy. ReLU at 0.1 also learned strongly; Sigmoid remained weak within eight epochs. Increasing the learning rate helped several activations, but Leaky ReLU at 0.1 became numerically unstable. A larger step size is therefore not universally better. Three byte-identical Training images were removed before splitting. No image cap was applied; every remaining single-fruit image in the pinned snapshot was used. Eight epochs provide a time-bounded comparison, not proof of asymptotic convergence.

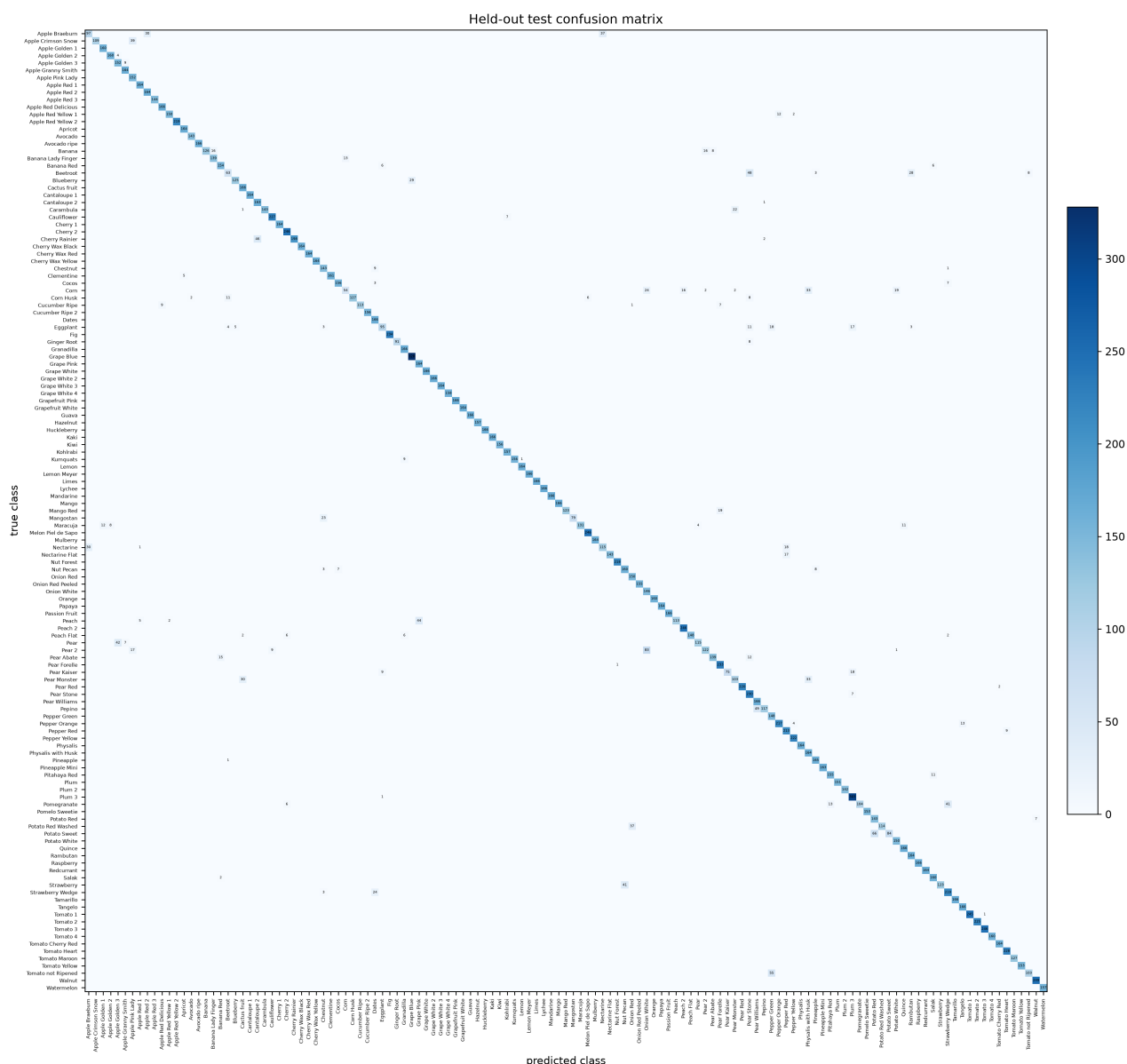
References and reproduction

Build: `python scripts/build_expanded_submission.py`. Training rerun: `python experiments/train_expanded.py --run`. The notebook generated beside this report reads the same evidence and can invoke that training command.

[1] H. Muresan and M. Oltean, Fruit recognition from images using deep learning, *Acta Universitatis Sapientiae, Informatica*, 10(1), 2018. <https://arxiv.org/abs/1712.00580>

[2] Fruits-360 dataset repository, Horea94/Fruit-Images-Dataset. <https://github.com/Horea94/Fruit-Images-Dataset>

Appendix A — Full held-out test confusion matrix



Rows are true classes; columns are predicted classes; exact label order is scope.classes in summary.json.